

Unit 6: Cubic & Cube Root Functions

Test Review: Transformations & Inverses

Name: _____

Period: _____

Part 1: Quick Concepts

1. A function and its inverse are reflections across the line _____.

2. If $f(7) = 20$, then $f^{-1}(20) =$ _____.3. If $f(-5) = 12$, what point is on $f^{-1}(x)$? _____4. If $f(6) = -3$, what point is on $f(x)$? _____

Part 2: Identify Parameters & Describe Transformations

5. $f(x) = -3(x + 2)^3 - 4$

$a =$ _____ $h =$ _____ $k =$ _____

Transformations:

6. $f(x) = \frac{1}{2}\sqrt[3]{x - 5} + 3$

$a =$ _____ $h =$ _____ $k =$ _____

Transformations:

Part 3: Write Equations from Descriptions

7. $f(x) = x^3$ is reflected over the x-axis, shifted left 4, and up 6.

$g(x) =$ _____

8. $f(x) = \sqrt[3]{x}$ is vertically stretched by 2, shifted right 3, and down 1.

$g(x) =$ _____

9. The function $g(x) = (x - c)^3 + d$ has its center point at $(-3, 7)$.

$c =$ _____ $d =$ _____

Part 4: Graph Transformations Find the center, transform the anchor points, graph, and find intercepts.

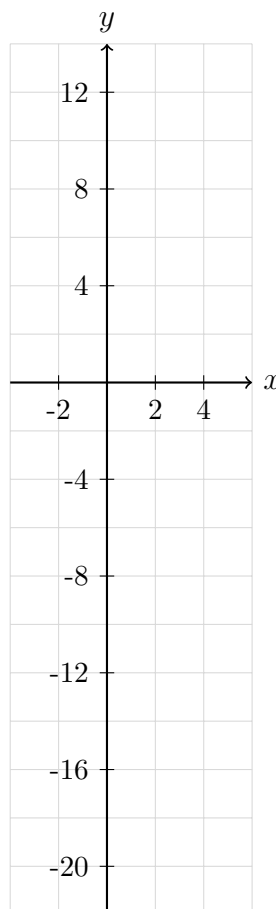
10. $f(x) = 2(x - 1)^3 - 4$

$a = \underline{\hspace{1cm}}$ $h = \underline{\hspace{1cm}}$ $k = \underline{\hspace{1cm}}$

Center: $(\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$

Parent	New
$(-2, -8)$	$(\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$
$(-1, -1)$	$(\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$
$(0, 0)$	$(\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$
$(1, 1)$	$(\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$
$(2, 8)$	$(\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$

y-intercept: $\underline{\hspace{1cm}}$ x-intercept: $\underline{\hspace{1cm}}$



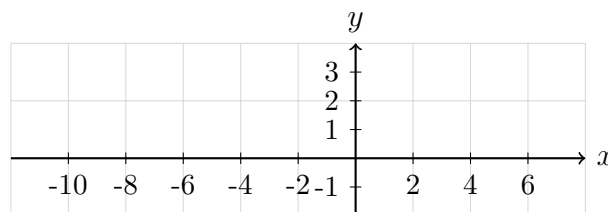
11. $f(x) = -\sqrt[3]{x+2} + 1$

$a = \underline{\hspace{1cm}}$ $h = \underline{\hspace{1cm}}$ $k = \underline{\hspace{1cm}}$

Center: $(\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$

Parent	New
$(-8, -2)$	$(\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$
$(-1, -1)$	$(\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$
$(0, 0)$	$(\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$
$(1, 1)$	$(\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$
$(8, 2)$	$(\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$

y-intercept: $\underline{\hspace{1cm}}$ x-intercept: $\underline{\hspace{1cm}}$



Part 5: Transform Specific Points

12. The parent function $f(x) = x^3$ contains the points $(1, 1)$ and $(-2, -8)$. The transformed function is $g(x) = -3(x - 2)^3 + 5$.

$$(1, 1) \rightarrow (\text{ }, \text{ }) \quad (-2, -8) \rightarrow (\text{ }, \text{ })$$

Find the x-intercept of $g(x)$: _____

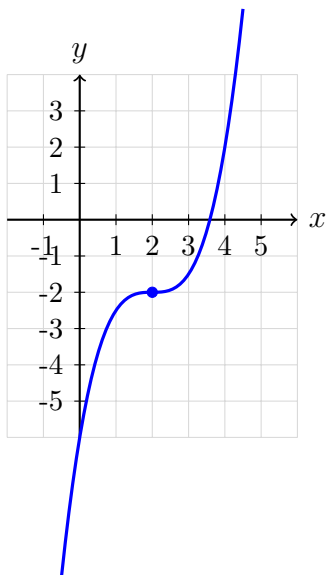
13. The parent function $f(x) = \sqrt[3]{x}$ contains the points $(8, 2)$ and $(-1, -1)$. The transformed function is $g(x) = -\sqrt[3]{x - 4} + 2$.

$$(8, 2) \rightarrow (\text{ }, \text{ }) \quad (-1, -1) \rightarrow (\text{ }, \text{ })$$

Find the x-intercept of $g(x)$: _____

Part 6: Write Equations from Graphs

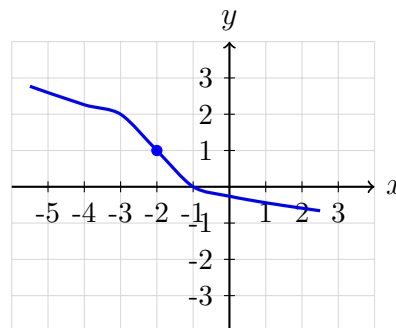
14.



Center: _____

$f(x) =$ _____

15.



Center: _____

$f(x) =$ _____

Part 7: Find Inverse Functions

16. Find the inverse of $f(x) = (x + 4)^3 - 2$

$$f^{-1}(x) = \underline{\hspace{2cm}}$$

17. Find the inverse of $f(x) = (2x - 1)^3 + 3$

$$f^{-1}(x) = \underline{\hspace{2cm}}$$

Part 8: Error Analysis

18. Marcus was asked to shift $f(x) = x^3$ up 4 units and right 2 units. He wrote $g(x) = (x + 2)^3 + 4$. Explain his error:

Correct equation: $g(x) = \underline{\hspace{2cm}}$

19. Destiny says $f(x) = 3(x - 1)^3 + 2$ is vertically compressed. Find and correct her error:

Part 9: Multiple Choice

20. Which represents $f(x) = x^3$ reflected over the x-axis and shifted left 5?

A. $g(x) = -x^3 + 5$ B. $g(x) = -x^3 - 5$ C. $g(x) = -(x - 5)^3$ D. $g(x) = -(x + 5)^3$

21. Given $h(x) = x^3$ and $p(x) = -2(x - 3)^3 + 1$, which describes the transformations?

- A. Vertical stretch by 2, reflect over x-axis, right 3, up 1
- B. Vertical stretch by 2, reflect over x-axis, left 3, up 1
- C. Vertical compress by $\frac{1}{2}$, reflect over x-axis, right 3, down 1
- D. Vertical stretch by 2, right 3, down 1

22. In $f(x) = 4(x - 2)^3 + 5$:

Vertical stretch: _____

Vertical shift: _____

Horizontal shift: _____

Part 10: Finding Intercepts

23. For the function $g(x) = 2(x + 1)^3 - 16$:

(a) Find the y-intercept by evaluating $g(0)$.

y-intercept: _____

(b) Find the x-intercept by setting $g(x) = 0$ and solving.

x-intercept: _____

Part 11: Verify Inverses by Composition (Advanced)

Show that $f(f^{-1}(x)) = x$ OR $f^{-1}(f(x)) = x$ to verify the functions are inverses.

24. Verify that $f(x) = (x - 1)^3 + 2$ and $f^{-1}(x) = \sqrt[3]{x - 2} + 1$ are inverses by computing $f(f^{-1}(x))$:

25. Given $f(x) = (2x + 3)^3 - 4$:

(a) Find $f^{-1}(x)$:

(b) Verify by showing $f^{-1}(f(x)) = x$: